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解: (1)
$$\begin{cases} u_t = u_{xx} & x \in \mathbb{R}; t > 0. \\ u|_{t=0} = \varphi(x) \end{cases}$$

(2) 积分二次得到: $u(x, y) = \frac{1}{4} x^2 y^2 + f(x) + g(y)$; 其中 f, g 为 γ 可微函数.

(3). ①. $f(t, x) = 0$ 时, 直接使用 d'Alembert 公式

$$\begin{aligned} u(t, x) &= \frac{(x - \sqrt{2}t) + (x + \sqrt{2}t)}{2} + \frac{1}{2\sqrt{2}} \int_{x - \sqrt{2}t}^{x + \sqrt{2}t} 2 \sin \xi \, d\xi \\ &= x + \frac{1}{\sqrt{2}} (\cos(x - \sqrt{2}t) - \cos(x + \sqrt{2}t)). \end{aligned}$$

② $f(t, x) = 3e^t \sin x$ 时, 令 $u = u_1 + u_2$.

其中 u_1 满足
$$\begin{cases} u_{1t} = 2u_{1xx} + f(t, x). \\ u_1|_{t=0} = 0. \end{cases}$$

u_2 满足
$$\begin{cases} u_{2t} = 2u_{2xx} \\ u_2|_{t=0} = x. \quad u_{2t}|_{t=0} = 2\sin x \end{cases}$$

则 u_2 即为 ① 中的解.

对于 u_1 , 由齐次化原理, 考虑方程
$$\begin{cases} w_t = 2w_{xx} \\ w|_{t=0} = 0. \quad w|_{t=\tau} = f(\tau, x) = 3e^\tau \sin x \end{cases}$$

$$\begin{aligned} \text{则 } w(t, x; \tau) &= \frac{1}{2\sqrt{2}} \int_{x - \sqrt{2}(t-\tau)}^{x + \sqrt{2}(t-\tau)} 3e^\tau \sin \xi \, d\xi \\ &= \frac{3e^\tau}{\sqrt{2}} \sin x \sin[\sqrt{2}(t-\tau)]. \end{aligned}$$

$$\Rightarrow u_1 = \int_0^t \frac{3e^\tau}{\sqrt{2}} \sin x \sin[\sqrt{2}(t-\tau)] \, d\tau.$$

$$u = u_1 + u_2 = \int_0^t \frac{3e^\tau}{\sqrt{2}} \sin x \sin[\sqrt{2}(t-\tau)] \, d\tau + x + \frac{1}{\sqrt{2}} \sin x \sin \sqrt{2}t.$$

二. 解: (1) 设 $Y_n(x) = \cos kx \Rightarrow sk = (n + \frac{1}{2})\pi \Rightarrow k = \frac{(n + \frac{1}{2})\pi}{s}$

$\Rightarrow Y_n(x) = \cos \frac{(n + \frac{1}{2})\pi x}{s}$. $\lambda_n = \left(\frac{(n + \frac{1}{2})\pi}{s} \right)^2$ $n=0, 1, 2, \dots$

(2) 对于 $f(x) = 1$. 展开为 $f(x) = \sum_{k=0}^{\infty} C_k Y_k(x)$.

其中 $C_k = \frac{1}{\|Y_k(x)\|^2} \int_0^s f(x) C_k(x) dx = \frac{s+1}{(n + \frac{1}{2})\pi} \cdot \frac{2}{s} = \frac{(-1)^n \cdot 2}{(n + \frac{1}{2})\pi}$

(2) 设 $U(t, x) = T(t) X(x)$. 代入波动方程.

$T''(t) X(x) = T(t) X''(x) + \pi^2 T(t) X(x) \Rightarrow \frac{T''(t)}{T(t)} = \frac{X''(x)}{X(x)} + \pi^2 \triangleq \lambda$. ①

得到特征值问题: $\begin{cases} X''(x) - \lambda X(x) = 0. \\ X'(0) = X'(1) = 0. \end{cases}$

$\Rightarrow X_n(x) = \cos(n\pi x)$. $\lambda_n = -(n\pi)^2$. $(n=0, 1, 2, \dots)$ 代入①得:

$T''(t) + (n^2 - 1)\pi^2 T(t) = 0 \Rightarrow T_n(t) = \begin{cases} \sin \sqrt{n^2 - 1} \pi t + \cos \sqrt{n^2 - 1} \pi t, & n \geq 2 \\ e^{\pi t} + e^{-\pi t}, & n=0 \\ At + B, & n=1. \end{cases}$

$(A_0 e^{\pi t} + B_0 e^{-\pi t}) + (A_1 t + B_1) \cos \pi x + \sum_{n=2}^{\infty} [A_n \cos \sqrt{n^2 - 1} \pi t + B_n \sin \sqrt{n^2 - 1} \pi t] \cos n\pi x$

由 $U(0, x) = \cos \pi x \Rightarrow (A_0 + B_0) + B_1 \cos \pi x + \sum_{n=2}^{\infty} A_n \cos n\pi x = \cos \pi x$.

$\Rightarrow \begin{cases} B_1 = 1 \\ A_n = 0 \ (n \geq 2) \\ A_0 + B_0 = 0 \end{cases} \Rightarrow U(t, x) = (A_0 e^{\pi t} - A_0 e^{-\pi t}) + (A_1 t + 1) \cos \pi x + \sum_{n=2}^{\infty} B_n \sin \sqrt{n^2 - 1} \pi t \cos n\pi x$

$\Rightarrow U_t(0, x) = 2\pi A_0 + A_1 \cos \pi x + \sum_{n=2}^{\infty} B_n \sqrt{n^2 - 1} \pi \cos(n\pi x) = 1 + \cos 2\pi x$

$\Rightarrow \begin{cases} A_1 = 0 \\ A_0 = \frac{1}{2\pi} \\ B_2 = \frac{1}{\sqrt{3}\pi} \\ B_n = 0 \ (n \geq 3) \end{cases}$ 综上: $U(t, x) = \frac{1}{2\pi} (e^{\pi t} - e^{-\pi t}) + \cos \pi x + \frac{1}{\sqrt{3}\pi} \sin \sqrt{3}\pi t \cos 2\pi x$. $\square$



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二. 解: 做极坐标变换. 由对称性可化为:
$$\begin{cases} u_t = u_{rr} + \frac{1}{r} u_r & (r \in [0, 2], t > 1) \\ u|_{r=2} = 0 \\ u|_{t=1} = 4 \end{cases}$$

设 $u(t, r) = T(t) R(r)$.

$$\Rightarrow \frac{T'(t)}{T(t)} = \frac{R''(r)}{R(r)} + \frac{R'(r)}{rR(r)} \triangleq -\lambda \quad (1)$$

$$\Rightarrow \begin{cases} rR''(r) + R'(r) + \lambda r R(r) = 0 & \text{为0阶 Bessel 方程.} \\ R(2) = 0. \end{cases}$$

$\Rightarrow R(r) = J_0(w_n r)$. $\lambda_n = w_n^2$. 其中 $\{w_n\}$ 为方程 $J_0(w_2) = 0$ 的正实根. ($n \geq 1$)

将 λ_n 代入 (1) 得到: $T'(t) + w_n^2 T(t) = 0 \Rightarrow T_n(t) = e^{-w_n^2 t}$

$\Rightarrow$ 形式解为: $u(t, r) = \sum_{n=1}^{+\infty} A_n e^{-w_n^2 t} J_0(w_n r)$.

$$\text{代入 } t=1 \Rightarrow \sum_{n=1}^{+\infty} A_n e^{-w_n^2} J_0(w_n r) = 4.$$

将 $4 \equiv y$ 在 $\{J_0(w_n r)\}$ 下展开. $\Rightarrow A_n e^{-w_n^2} \|J_0(w_n r)\|^2 = \int_0^2 4x J_0(w_n x) dx \quad (2)$

$$\text{对于 (2). 令 } w_n x = t \Rightarrow \int_0^{2w_n} \frac{4}{w_n} t J_0(t) dt = \frac{4}{w_n} t J_1(t) \Big|_0^{2w_n}$$

$$\text{而 } \|J_0(w_n r)\|^2 = 2 J_1^2(2w_n) = 8 J_1^2(w_n t)$$

$$\Rightarrow A_n = \frac{8 J_1(w_n t) e^{w_n^2}}{2 J_1^2(2w_n)}$$

$$\Rightarrow \text{形式解} = \sum_{n=1}^{+\infty} \frac{4 J_1(w_n t) e^{w_n^2}}{J_1^2(2w_n)} e^{-w_n^2 t} J_0(w_n r). \quad \square$$

四. 解: $\Delta_3 u = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial u}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial u}{\partial \theta}) = 0$

设 $U = R(r) \Theta(\theta)$. 代入拉普拉斯方程.

$$\Rightarrow \left\{ \begin{array}{l} \frac{1}{R(r)} \frac{d}{dr} (r^2 \frac{dR}{dr}) = \lambda \quad ① \\ \frac{1}{\Theta(\theta) \sin \theta} \frac{d}{d\theta} (\sin \theta \frac{d\Theta}{d\theta}) = -\lambda \quad ② \end{array} \right.$$

对于②为勒让德方程. 特征值 $\lambda_n = n(n+1)$. 特征函数 $\Theta_n(\theta) = P_n(\cos \theta)$.

λ_n 代入① $\Rightarrow R_n(r) = r^n + r^{-(n+1)}$

$\Rightarrow$ 形式解为: $U = \sum_{n=0}^{+\infty} (A_n r^n + B_n r^{-(n+1)}) P_n(\cos \theta)$.

由 $\lim_{r \rightarrow \infty} U = 1 \Rightarrow A_0 = 1, A_n = 0 (n \geq 1)$. 形式解为 $U = 1 + \sum_{n=0}^{+\infty} \frac{B_n}{r^{n+1}} P_n(\cos \theta)$

$\Rightarrow U_r = - \sum_{n=0}^{+\infty} \frac{(n+1) B_n}{r^{n+2}} P_n(\cos \theta) \Rightarrow U_r|_{r=2} = - \sum_{n=0}^{+\infty} \frac{(n+1) B_n}{2^{n+2}} P_n(\cos \theta) = 1 + \cos \theta + \cos^2 \theta$

$P_0(\cos \theta) = 1, P_1(\cos \theta) = \cos \theta, P_2(\cos \theta) = \frac{1}{2} (3 \cos^2 \theta - 1)$

$\Rightarrow -\frac{B_0}{4} + \frac{-2B_1}{8} \cos \theta + \frac{-3B_2}{16} \cdot \frac{1}{2} (3 \cos^2 \theta - 1) = 1 + \cos \theta + \cos^2 \theta$

$\Rightarrow B_2 = \frac{-32}{9}, B_1 = -4, B_0 = -\frac{16}{5}$

$\Rightarrow U = 1 + \frac{-16}{5} \cdot \frac{1}{r} P_0(\cos \theta) + \frac{-4}{r^2} P_1(\cos \theta) + \frac{-32}{9} \cdot \frac{1}{r^3} P_2(\cos \theta)$ a



五. 解: (1) $\varphi(x) = \delta(x)$ 时. 做 Fourier 变换, $\bar{u}(t, \lambda) = \int_{\mathbb{R}} u(t, x) e^{i\lambda x} dx$.

$$\Rightarrow \begin{cases} \frac{d\bar{u}}{dt} = -4\lambda^2 \bar{u} - i\lambda \bar{u} + \bar{u} = \bar{u}(1 - 4\lambda^2 - i\lambda) \\ \bar{u}|_{t=0} = 1. \end{cases}$$

$$\Rightarrow \bar{u} = e^{(1-4\lambda^2-i\lambda)t}$$

做 Fourier 逆变化: $u = \frac{1}{2\pi} \int_{\mathbb{R}} e^{(1-4\lambda^2-i\lambda)t} e^{-i\lambda x} d\lambda$.

$$= \frac{e^t}{4\sqrt{\pi t}} e^{-\frac{(x-t)^2}{16t}}$$

$$= \frac{1}{2} e^{\frac{15}{16}t} \int_{\mathbb{R}} \frac{1}{\sqrt{\pi}} e^{-(2\lambda + \frac{i}{4})^2 t} e^{-i\lambda x} d\lambda$$

(2) $\varphi(x) = \sin x$ 时

$$= \frac{1}{2} e^{\frac{15}{16}t} \int_{\mathbb{R}} \frac{1}{\sqrt{\pi}} e^{-(2\lambda + \frac{i}{4})^2 t} e^{-i\lambda x} d\lambda$$

$$\begin{aligned} \bar{\varphi}(x) &= \int_{\mathbb{R}} \sin x e^{i\lambda x} dx = \frac{1}{2i} \int_{\mathbb{R}} e^{i\pi(\lambda+1)} - e^{i\pi(\lambda-1)} d\lambda \\ &= \left(\delta(\lambda+1) - \delta(\lambda-1) \right) \frac{\pi}{i} \end{aligned}$$

$$\Rightarrow \bar{u} = \frac{\pi}{i} \left(\delta(\lambda+1) - \delta(\lambda-1) \right) e^{(1-4\lambda^2-i\lambda)t}$$

$$= \frac{\pi}{i} \left[\delta(\lambda+1) e^{(1-4-i)t} - \delta(\lambda-1) e^{(1-4-i)t} \right]$$

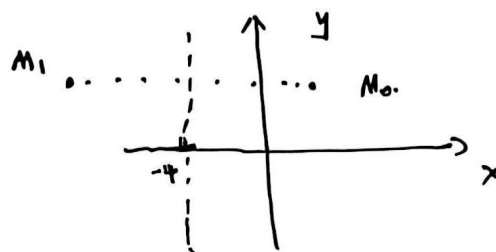
$$= \frac{\pi}{i} \left\{ \delta(\lambda+1) e^{(i-3)t} - \delta(\lambda-1) e^{(-i-3)t} \right\}$$

$$\Rightarrow u = \frac{1}{2\pi} \frac{\pi}{i} \int_{\mathbb{R}} \left(\delta(\lambda+1) e^{(i-3)t} - \delta(\lambda-1) e^{(-i-3)t} \right) e^{-i\lambda x} d\lambda$$

$$= e^{-3t} \sin(x+t).$$

六. 解: (1) 格林函数 G 满足

$$\begin{cases} \Delta G = -\delta(x-3, y-y) \\ G|_{x=-4} = 0. \end{cases}$$



记 $M_0(3, y)$ 为 Ω 中 $\forall$ - 点.

其关于 $x = -4$ 对称点为 $M_1(m, n)$.

其中 $n = y, m = -8 - 3$.

则 $G = \frac{1}{2\pi} \ln \frac{r(M, M_0)}{r(M, M_1)} + \ln r(M, M_1)$ 其中 $M(x, y)$.

(2). 令 $\begin{cases} x=2\beta \\ y=3\eta \end{cases} \Rightarrow$ 方程化为: $\begin{cases} u_{\beta\beta} + u_{\eta\eta} = \delta(2\beta, 3\eta) = \frac{1}{6} \delta(3\eta) = \Delta U \\ u|_{\partial\Omega} = \varphi(3\eta). \end{cases}$

令 $u = u_1 + u_2$. 其中 u_1 满足 $\begin{cases} \Delta u_1 = 0 \\ u_1|_{\partial\Omega} = \varphi(3\eta). \end{cases}$ u_2 满足 $\begin{cases} \Delta u_2 = \frac{1}{6} \delta(3\eta) \\ u_2|_{\partial\Omega} = 0 \end{cases}$

对于 u_1 , 当 $(x, y) \in \Omega$ 时.

$(3, y) \in \{(3, y) | 3 > -8, y \in \mathbb{R}\} = \Omega'$ 与 (1) 类似. 可求出该问题的 Green 函数 G_1 .

则 $u_1 = -\int_{\mathbb{R}} \varphi(3\eta) \frac{\partial G_1(3, y; \alpha, \beta)}{\partial n} d\eta$. 其中 $\mathbb{R}$ 为 $x = -4, M(3, y), M_0(\alpha, \beta)$.

对于 u_2 , 由 (1) 知 $u_2 = \frac{1}{6} G_1$. 其中 G_1 为 Ω' 中求得的格林函数.

综上: $u = u_1 + u_2 = \frac{1}{6} G_1 - \int_{\mathbb{R}} \varphi(3\eta) \frac{\partial G_1(3, y; \alpha, \beta)}{\partial n} d\eta$. $\square$